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Professor Yamir Moreno
Editor-in-Chief
Journal of Complex Networks

Dear Professor Moreno,

Please consider the manuscript "When does a spectral prior help graph learning? Connectivity-loss estimation under road-network disruptions" as an original research article in the *Journal of Complex Networks*.

The paper studies a structural problem at the intersection of network spectra, spatial infrastructure networks, and graph machine learning. Its central contribution is a hybrid estimator that learns only the finite-perturbation residual of an interpretable Fiedler edge-sensitivity approximation. The design is evaluated with GCN, GraphSAGE, and edge-aware MPNN backbones under independent, spatially correlated, and targeted-betweenness edge failures, with graph-disjoint synthetic tests, 13 OpenStreetMap areas in six countries, area-blocked and leave-one-country-out transfer, architecture ablations, area/country-clustered hierarchical intervals, eigengap and second-order controls, correction diagnostics, and scaling to 20,000 nodes. The main question is when an analytical prior helps under distribution shift and when it becomes a biased anchor. A reliability-context extension that failed to transfer is reported transparently as a negative ablation.

The manuscript fits the journal's interests in mathematical and numerical analysis of networks, network spectra, structural analysis, spatial networks, infrastructural networks, and AI for network complexity. The code, cached graph instances and compact result tables are available at github.com/lee-vtruong/ResiliRoad.

This manuscript is not being submitted elsewhere. Any related VMS60 poster or abstract will be disclosed in the submission record; the journal manuscript is substantially extended by multi-area transfer, controlled scaling, calibration, negative-result analysis and journal-specific discussion. Generative AI tools were used for language editing and software-development assistance. The author designed the study, checked the implementation and results, and accepts full responsibility for the submitted work.

Thank you for your consideration.

Sincerely,

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